

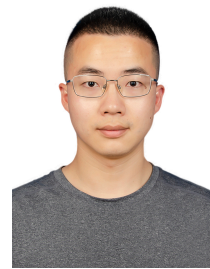
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Academic Qualifications

- 2022 – Present Ph.D. candidate in School of Mathematics, Southwest Jiaotong University
2019 – 2022 M.Sc. in School of Mathematics, Southwest Jiaotong University
2013 – 2017 B.Sc. in School of Mathematics and Information, China West Normal University

List of Publications

Publications During M.Sc.

- [1] **Z. Li**, H. Yan, “Differential spectra of a class of power permutations with Niho exponents,” *Adv. Math. Commun.*, vol. 17, no. 6, pp. 1468-1475, 2023.
- [2] **Z. Li**, C. Fan, W. Su, Y. Qi, “Aperiodic/periodic complementary sequence pairs over quaternions,” *Adv. Math. Commun.*, vol. 17, no. 6, pp. 1493-1506, 2023.
- [3] H. Yan, **Z. Li**, Z. Song, R. Feng, “Two classes of power mappings with boomerang uniformity 2,” *Adv. Math. Commun.*, vol. 16, no. 4, pp. 1111-1120, 2022.
- [4] H. Yan, **Z. Li**, “A note on the differential spectrum of a class of power mappings with Niho exponent,” *Cryptogr. Commun.*, vol. 14, no. 5, pp. 1081-1089, 2022.
- [5] **Z. Li**, H. Yan, D. Han, “A class of power functions with four-valued Walsh transform and related cyclic codes,” *Finite Fields Their Appl.*, vol. 83, pp. 102078, 2022.

Publications During Ph.D.

- [1] **Z. Li**, X. He, X. Tang, “Marker+codeword+marker: a coding structure for segmented single-insdel/single-edit channels,” to be submitted.

In this manuscript, we introduce binary marker-based VT code modification to achieve the capability of correcting segmented edit errors. We reproduce the best-known code redundancy for segmented insdel channels, contrasting our advantage in having linear-time encoder/decoder. Additionally, we present code construction for segmented insdel/substitution channels. Compared to the insdel scenario, the code redundancy increases by 3 bits while still maintaining a linear-time encoding/decoding scheme.

Future Research Topics

- Efficient Constrained codes for DNA storage systems
- Error Correcting Codes for Deletions and Other Reasonable Errors